Choose the required accuracy

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: __________________

1. Read the model: Round 384,650 to the nearest 10,000 and then to the nearest 1,000. Copy or complete the first step.

2. Round 2,749,501 to the nearest 100,000.

3. Round 999,499 to the nearest 1,000.

4. Miro says: Miro rounds to the largest place value he can see instead of the place named. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Round 384,650 to the nearest 10,000 and then to the nearest 1,000. Copy or complete the first step.
   Answer 380,000 and 385,000.

2. Round 2,749,501 to the nearest 100,000.
   Answer 2,700,000.

3. Round 999,499 to the nearest 1,000.
   Answer 999,000.

4. Miro says: Miro rounds to the largest place value he can see instead of the place named. Is this correct? Explain.
   Answer The question controls the accuracy. Mark that place before deciding.

Choose the required accuracy

Work independently. Show a complete method and check at least one answer.

Name: ____ Date: ________________

1. Round 384,650 to the nearest 10,000 and then to the nearest 1,000.

2. Round 2,749,501 to the nearest 100,000.

3. Round 999,499 to the nearest 1,000.

4. Solve this independently and show every stage: Round 384,650 to the nearest 10,000 and then to the nearest 1,000.

5. Use a second method or representation for: Round 2,749,501 to the nearest 100,000.

6. Check this result using an inverse, estimate or known fact: Round 999,499 to the nearest 1,000.

7. A number rounds to 730,000 to the nearest 10,000. State its full possible range.

8. Identify and correct this misconception: Miro rounds to the largest place value he can see instead of the place named.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Round 999,499 to the nearest 1,000.

Teacher answers and checking prompts

1. Round 384,650 to the nearest 10,000 and then to the nearest 1,000.
   Answer 380,000 and 385,000.

2. Round 2,749,501 to the nearest 100,000.
   Answer 2,700,000.

3. Round 999,499 to the nearest 1,000.
   Answer 999,000.

4. Solve this independently and show every stage: Round 384,650 to the nearest 10,000 and then to the nearest 1,000.
   Answer 380,000 and 385,000.

5. Use a second method or representation for: Round 2,749,501 to the nearest 100,000.
   Answer Expected result: 2,700,000. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Round 999,499 to the nearest 1,000.
   Answer Expected result: 999,000. A valid check must be shown.

7. A number rounds to 730,000 to the nearest 10,000. State its full possible range.
   Answer 725,000 to 734,999.

8. Identify and correct this misconception: Miro rounds to the largest place value he can see instead of the place named.
   Answer The question controls the accuracy. Mark that place before deciding.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Round 999,499 to the nearest 1,000.
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Choose the required accuracy

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ______________________________ Date: ____________

1. A number rounds to 730,000 to the nearest 10,000. State its full possible range.

2. Prove the answer to this question, rather than only calculating it: Round 384,650 to the nearest 10,000 and then to the nearest 1,000.

3. Solve this in two different ways and compare them: Round 2,749,501 to the nearest 100,000.

4. Work backwards from the answer to reconstruct the question: 999,000.

5. Explain why this reasoning fails: Miro rounds to the largest place value he can see instead of the place named.

6. Create a new example that tests the same idea as: Round 999,499 to the nearest 1,000.

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. A number rounds to 730,000 to the nearest 10,000. State its full possible range.
   Answer 725,000 to 734,999.

2. Prove the answer to this question, rather than only calculating it: Round 384,650 to the nearest 10,000 and then to the nearest 1,000.
   Answer Expected result: 380,000 and 385,000. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: Round 2,749,501 to the nearest 100,000.
   Answer Expected result: 2,700,000. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 999,000.
   Answer One valid reconstruction is: Round 999,499 to the nearest 1,000. Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro rounds to the largest place value he can see instead of the place named.
   Answer The question controls the accuracy. Mark that place before deciding.

6. Create a new example that tests the same idea as: Round 999,499 to the nearest 1,000.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.